🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 5 · STANDARD 6.GR.1

Determine the Area of Trapezoids

Lesson 5-3 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can find the area of a trapezoid using the formula $A = \frac{1}{2}(b_1 + b_2) \times h$.

Language Objective: I can explain my steps using the words trapezoid, base, height, and area.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Area of Trapezoids | **Trapezoid** | **Parallel** | **Base 1 (b1)** | **Base 2 (b2)** | **Height** | **Perpendicular** | **Area**

👆 *Tap any word to see what it means and a picture.*

Level 3: try the blanks from memory first, then check the bank.

1. The space inside a trapezoid, found with $A = \frac{1}{2}(b_1 + b_2)h$, is the ____________________.

2. A quadrilateral with exactly one pair of parallel sides is a ____________________.

3. Two sides that stay the same distance apart and never meet are ____________________.

4. One of the two parallel sides of a trapezoid is ____________________.

5. The other parallel side of a trapezoid is ____________________.

6 The perpendicular distance between the two parallel bases is the ______.

7 Two lines that meet at a 90° square corner are ______.

8 The amount of flat space inside the trapezoid is its ______.

Write About the Math **Guided ESOL writing**

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

How many bags of concrete does the builder need?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- [ ] **Area of Trapezoids** — The space inside a trapezoid. Formula: $A = 1/2(b_1 + b_2)h$.
  Área de trapecios — El espacio dentro de un trapecio. Fórmula: $A = 1/2(b_1 + b_2)h$.
- [ ] **Trapezoid** — A four-sided shape with just one pair of parallel sides.
  Trapecio — Una figura de cuatro lados con solo un par de lados paralelos.
- [ ] **Parallel** — Two lines or sides that stay the same distance apart and never meet.
  Paralelas — Dos líneas o lados que se mantienen a la misma distancia y nunca se cruzan.
- [ ] **Base 1 (b1)** — One of the two parallel sides of a trapezoid.
  Base 1 (b1) — Uno de los dos lados paralelos de un trapecio.
- [ ] **Base 2 (b2)** — The other parallel side of a trapezoid.
  Base 2 (b2) — El otro lado paralelo de un trapecio.

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

The driveway's area is ____ sq ft because A = 1/2 x (____ + ____) x ____. He needs ____ bags because ____ / 6 = ____.

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: We use both bases because the top and bottom are different lengths.

Evidence: Students explain a trapezoid has two parallel sides of different lengths, so using only one would over- or under-count the area; strong answers mention averaging the bases.

Reasoning: This evidence connects the definition of area of trapezoids to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The driveway's area is ____ sq ft because A = 1/2 x (____ + ____) x ____. He needs ____ bags because ____ / 6 = ____.**
- **My answer is ____.**
  Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Area of Trapezoids
2. **Notes 2:** Trapezoid
3. **Notes 3:** Parallel
4. **Notes 4:** Base 1 (b1)
5. **Notes 5:** Base 2 (b2)
6. **Notes 6:** Height
7. **Notes 7:** Perpendicular
8. **Notes 8:** Area
9. **Watch for this mistake:** *A common mistake in Area of Trapezoids is forgetting the ½ in $A = \frac{1}{2} \times (b1 + b2) \times h$. After adding the bases and multiplying by the height, students stop there instead of taking half — for example, with bases 6 ft and 10 ft and a height of 4 ft, they compute $(6 + 10) \times 4 = 64$ sq ft instead of taking half to get the correct 32 sq ft. That unhalved number is really the area of the parallelogram made from two trapezoids, not the area of one trapezoid. Before you submit, always ask: "Did I take half of my last answer?"*
10. **Try It 1:** A. 48 sq ft — *$A = \frac{1}{2} \times (9 + 7) \times 6 = \frac{1}{2} \times 16 \times 6 = \frac{1}{2} \times 96 = 48$ sq ft.*
11. **Try It 2:** A. 48 sq ft — *One trapezoid is half the parallelogram: $96 \div 2 = 48$ sq ft. This matches $A = \frac{1}{2} \times (5 + 11) \times 6 = \frac{1}{2} \times 16 \times 6 = 48$ sq ft — the ½ comes from the trapezoid being half of the parallelogram.*